Miscelánea Mecánica Analítica #2

24 de abril de 2026

Calculo directo de la cuasi-invarianza

$$\mathcal{L}(x,\dot{x}) = m\frac{\dot{x}^2}{2} + \frac{a}{x^2} \qquad \begin{cases} x \mapsto x' = x - \varepsilon t x \\ t \mapsto t' = t + \varepsilon t^2, \ (\varepsilon \ll 1) \end{cases} \quad \therefore \begin{cases} dx' = dx - \varepsilon(xdt + tdx) \\ dt' = dt + 2\varepsilon t dt \end{cases}$$

$$\frac{1}{1 \pm r\varepsilon t} := 1 \mp r\varepsilon t + \underbrace{\mathfrak{o}(\varepsilon^2)}_{\simeq 0} \ (r \in \mathbb{R})$$

$$\dot{x}' := \frac{dx'}{dt'} := \frac{(dx - \varepsilon x dt - \varepsilon t dx)}{dt + 2\varepsilon t dt} = \frac{(1-\varepsilon t)dx - \varepsilon x dt}{(1+2\varepsilon t)dt} = \frac{(1-\varepsilon t)}{(1+2\varepsilon t)}\dot{x} - \frac{\varepsilon x}{(1+2\varepsilon t)} =$$

$$= (1-2\varepsilon t)[(1-\varepsilon t)\dot{x} - \varepsilon x] \quad (\dot{x}')^2 := \boxed{(1-2\varepsilon t)^2[(1-\varepsilon t)\dot{x} - \varepsilon x]^2}$$

$$x' = (1+\varepsilon t)x \ \therefore (x')^2 = (1-\varepsilon t)^2 x^2 \quad \Rightarrow \frac{1}{(x')^2} = \frac{1}{(1-\varepsilon t)^2 x^2} := \boxed{\frac{1+2\varepsilon t}{x^2}}$$

$$\therefore \mathcal{L} \mapsto \mathcal{L}' := m\frac{\dot{x}'^2}{2} + a\frac{1}{x'^2} = \frac{m(1-2\varepsilon t)^2}{2}[(1-\varepsilon t)\dot{x} - \varepsilon x]^2 + \frac{a(1+2\varepsilon t)}{x^2}$$

$$\delta(\mathcal{L}dt) \equiv \lim_{\varepsilon \to 0} \frac{\mathcal{L}'dt' - \mathcal{L}dt}{\varepsilon} \leftarrow \Delta(\mathcal{L}dt) = \mathcal{L}'dt' - \mathcal{L}dt.$$

$$\mathcal{L}'dt' = \left[\frac{m(1-2\varepsilon t)^2}{2}[(1-\varepsilon t)\dot{x} - \varepsilon x]^2 + \frac{a(1+2\varepsilon t)}{x^2}\right](1+2\varepsilon t)dt :=$$

$$:= \left[\frac{m(1-2\varepsilon t)}{2}[(1-\varepsilon t)\dot{x} - \varepsilon x]^2 + (1+2\varepsilon t)^2 \underbrace{\frac{a}{x^2}}_{-V(x)}\right]dt =$$

$$= \frac{m}{2}[(1-\varepsilon t)\dot{x} - \varepsilon x]^2\, dt - m\varepsilon t[(1-\varepsilon t)\dot{x} - \varepsilon x]^2\, dt - \underbrace{(1+2\varepsilon t)^2}_{\simeq(1+4\varepsilon t)} V(x)dt$$

$$[(1-\varepsilon t)\dot{x} - \varepsilon x]^2 = (1-\varepsilon t)^2\dot{x}^2 - 2(1-\varepsilon t)\dot{x}\varepsilon x + \varepsilon^2 x^2 = (1 - 2\varepsilon t + \varepsilon^2 t^2)\dot{x}^2 - 2(\varepsilon - \varepsilon^2 t)x\dot{x} + \varepsilon^2 x^2 :=$$

$$:= (1-2\varepsilon t)\dot{x}^2 - 2\varepsilon x\dot{x} + \mathfrak{o}(\varepsilon^2)$$

$$\therefore \ \mathcal{L}'dt' := \{\frac{m}{2}[(1-2\varepsilon t)\dot{x}^2 - 2\varepsilon x\dot{x}] - m\varepsilon t[(1-2\varepsilon t)\dot{x}^2 - 2\varepsilon x\dot{x}] - (1+4\varepsilon t)V(x)\}dt$$

$$\frac{m}{2}[(1-2\varepsilon t)\dot{x}^2 - 2\varepsilon x\dot{x}] := T - m\varepsilon(t\dot{x}^2 + x\dot{x})$$

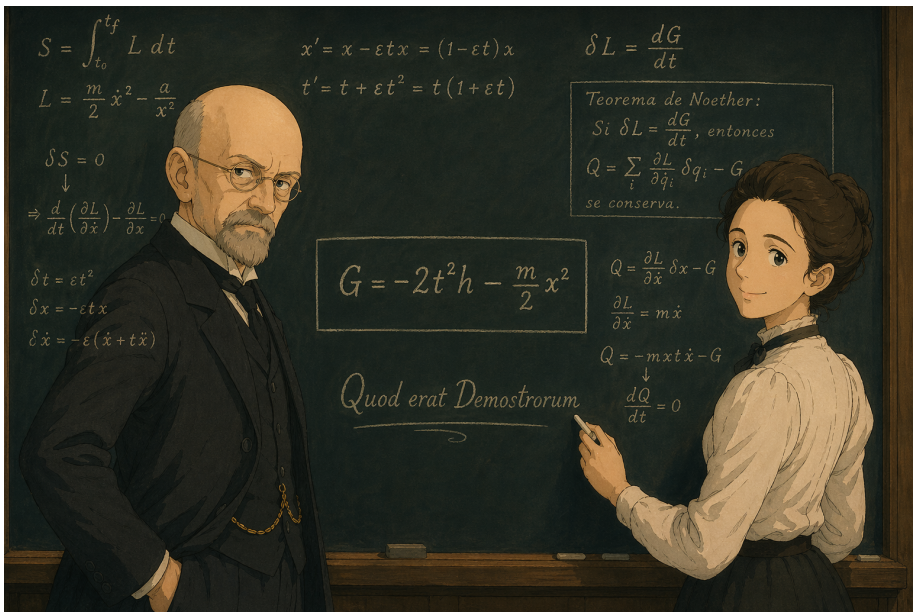
$$\Delta(\mathcal{L}dt) := \{-m\varepsilon(\dot{x}^2 t + x\dot{x}) - m\varepsilon t[(1-2\varepsilon t)\dot{x}^2 - 2\varepsilon x\dot{x}] - 4\varepsilon t V(x)\}dt$$

$$\therefore\ \delta(\mathcal{L}\,dt) = \lim_{\varepsilon\to 0}\frac{\Delta(\mathcal{L}dt)}{\varepsilon} := -\{m[(\dot{x}^2 t + x\dot{x}) + t\dot{x}^2]dt + 4tV(x)\}\,dt$$

$$\delta(\mathcal{L}\,dt) := t\mathcal{F}\,dt + \mathcal{D}\,dt := -t(2m\dot{x}^2 + 4V(x))\,dt - mx\dot{x}\,dt$$

$$h \equiv \mathcal{E} = T + V \qquad \therefore \delta(\mathcal{L}\,dt) := -4th - mx\dot{x} \Rightarrow \delta S = -\int_{t_0}^{t_f}(4th - mx\dot{x})\,dt$$

$$\therefore \boxed{\delta S := \int_{t_0}^{t_f}\frac{dG}{dt}dt \qquad \frac{dG}{dt} := -4th - mx\dot{x}}$$

$$\frac{dG}{dt} = -4th - mx\dot{x} \Rightarrow G = -4h\int tdt - m\int x\dot{x}dt := -2ht^2 - \frac{m}{2}x^2$$

$$\boxed{G = -2t^2h - \frac{m}{2}x^2}$$